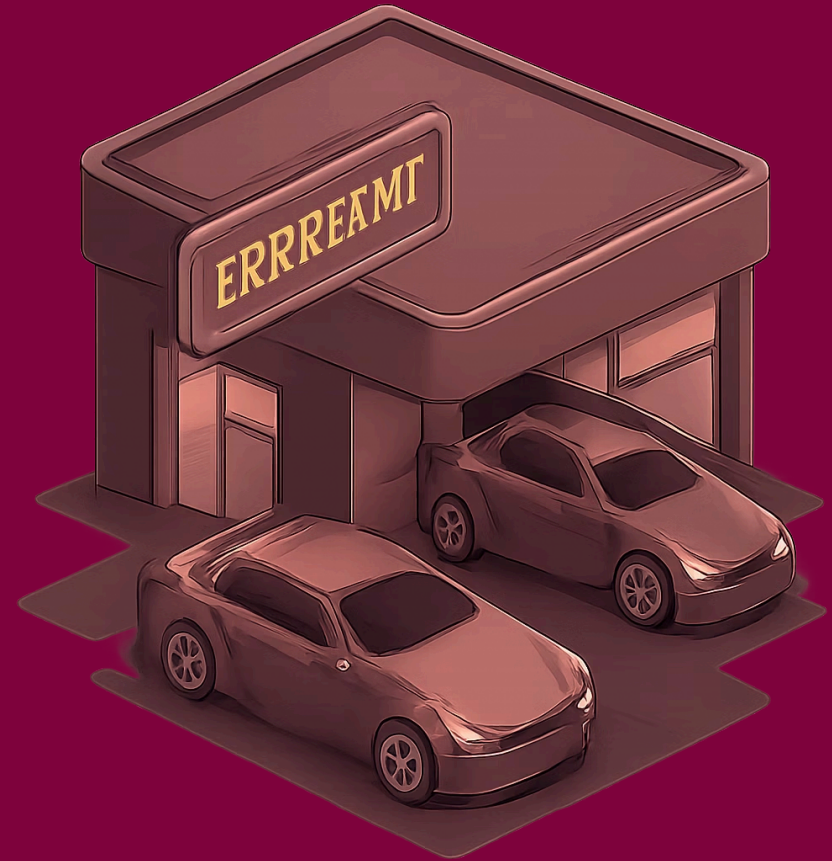


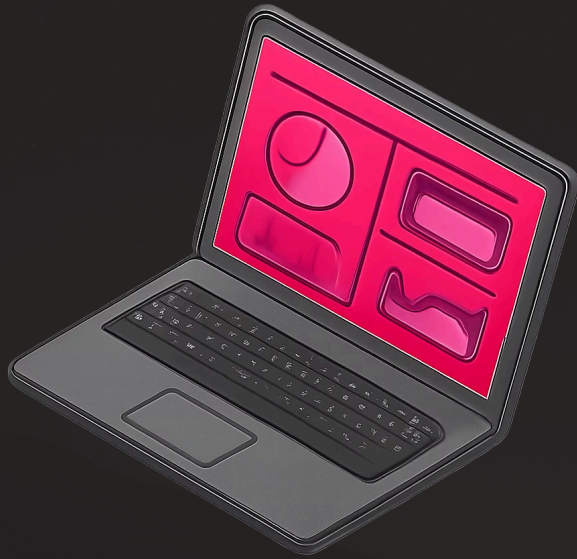
Car Rental Fleet Manager

Programming for Problem Solving – Java

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Abstract



A **Java-based console application** for managing rental vehicles, customers, and transactions end-to-end.

Core Operations

View, search, rent, and return cars with real-time availability tracking

Customer & Records

Manage customer information and maintain persistent rental records

Analytics & Reports

Sorting, filtering, receipt generation, and fleet statistics

Java Progression

Demonstrates Modules 1–6 of the Java PSP syllabus in one cohesive project

Project Objectives & Key Features



Fleet Management

Manage car info and track real-time availability across the fleet



Search & Filter

Quickly search and filter vehicles by brand, price, or availability



Transactions

Handle rental and return processing with automated cost calculation



Customer Records

Maintain customer details and complete rental histories



Fleet Statistics

Generate reports: available/rented count and total revenue summary



File Persistence

File-based record storage for data durability across sessions

Advantages

- Reduces manual work and paperwork
- Improves organization and data management
- Enables quick searching and filtering
- Automates cost calculation and billing
- Demonstrates modular and scalable design
- Supports efficient fleet analysis and reporting

Java PSP Modules Used

Module 1

Variables, data types, operators, Scanner, console I/O, algorithm design

Module 2

if-else, switch, for / while / do-while loops, break, continue

Module 3

Methods, arrays, searching, sorting, recursion, method overloading

Module 4

Classes, objects, constructors, encapsulation, inheritance, interfaces, polymorphism, exceptions

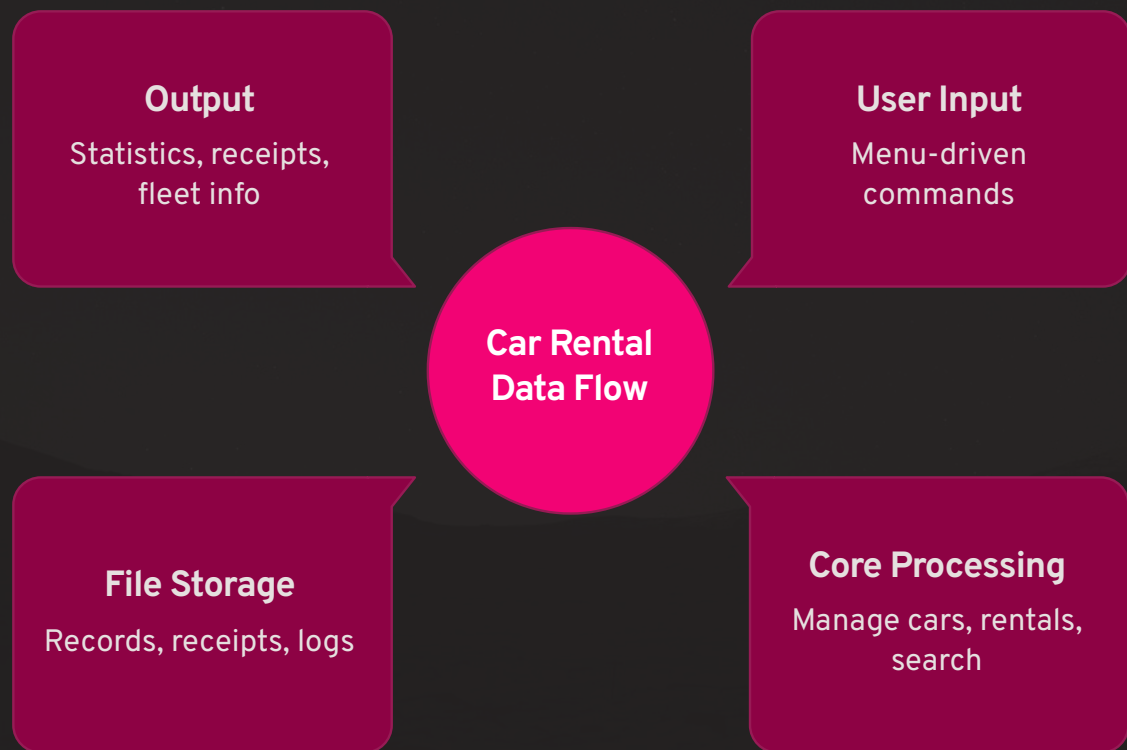
Module 5

Strings, file I/O, BufferedReader/Writer, Collections.sort(), Comparable, Comparator

Module 6

ArrayList, HashMap, HashSet, generics, lambda expressions, streams, Collectors

System Architecture & Class Design



Core Class Structure

1	Car carId, carName, brand, rentalPrice, available
2	Customer customerId, customerName, phoneNumber
3	Rental rentalId, customer, car, numberOfDays, totalCost
4	CarRentalManager Manages all lists; coordinates cars, customers & rental operations

Implementation Details – Modules 1 to 6

1

M1 – I/O

Input handling, variables, operators,
Scanner, algorithm design

2

M2 – Control

Conditional logic, switch menus,
loop-based iterations

3

M3 – Methods

Modular methods, arrays for storage,
searching & sorting algorithms

1

M4 – OOP

OOP principles, constructors,
encapsulation, inheritance,
exception handling

2

M5 – Strings/IO

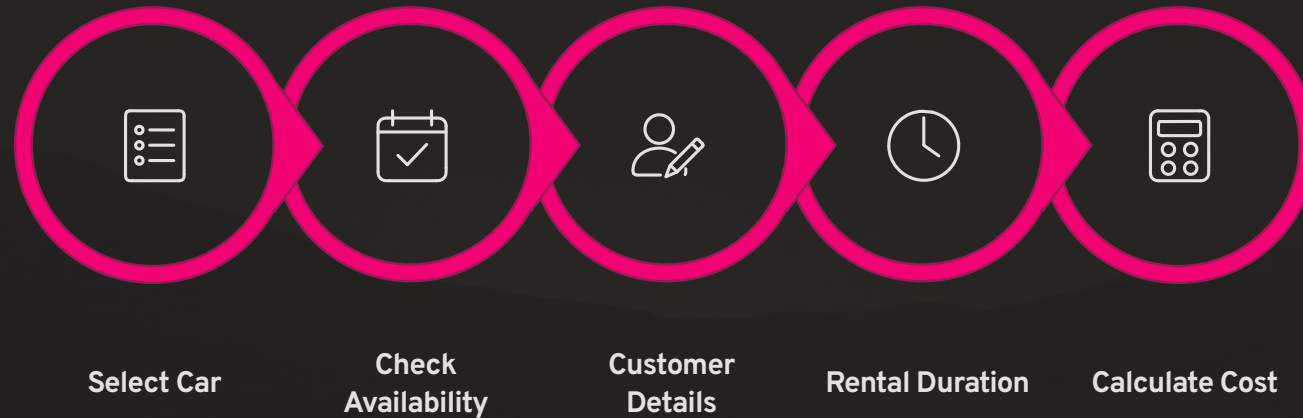
String operations,
BufferedReader/Writer for
persistence,
Comparable/Comparator sorting

3

M6 – Collections

ArrayList, HashMap, generics,
lambda expressions, stream
processing

Rental Process Workflow & Sample Output



OUTPUT SCREENSHOT 1

```
vscode>_19000\juc_ws\juc.js-java-project\src - CarManager
Enter your choice: 1

===== CAR FLEET =====
ID: 101 | Name: Swift | Category: Hatchback | Price/Day: ₹1500.0 | Available: true
ID: 102 | Name: Baleno | Category: Hatchback | Price/Day: ₹1700.0 | Available: true
ID: 103 | Name: Creta | Category: SUV | Price/Day: ₹2500.0 | Available: true
ID: 104 | Name: City | Category: Sedan | Price/Day: ₹2200.0 | Available: false
ID: 105 | Name: Nexon | Category: SUV | Price/Day: ₹1800.0 | Available: true

=====
CAR RENTAL FLEET MANAGER
=====
1. Display All Cars
2. Search Car by ID
3. Search Car by Name
4. Show Available Cars
5. Rent Car
6. Calculate Rental Cost
7. Sort Cars by Price
8. Count Available Cars
9. Calculate Average Price
```

OUTPUT SCREENSHOT 2

```
VSCODESWS_19666\jdt_ws\jdt.ls-java-project\bin Carmanager
0. CALL
```

```
=====
Enter your choice: 2
```

```
Enter Car ID: 104
```

```
ID: 104 | Name: City | Category: Sedan | Price/Day: ?2200.0 | Available: false
```

```
=====
CAR RENTAL FLEET MANAGER
=====
```

1. Display All Cars
2. Search Car by ID
3. Search Car by Name
4. Show Available Cars
5. Rent Car
6. Calculate Rental Cost
7. Sort Cars by Price
8. Count Available Cars
9. Calculate Average Price
10. Array Analysis
11. Save Fleet to File
12. Read Fleet from File
0. Exit



Sample Fleet Data

Car ID	Name	Category	Price / Day	Available
101	Swift	Hatchback	₹1500.0	✓ true
102	Baleno	Hatchback	₹1700.0	✓ true
103	Creta	SUV	₹2500.0	✓ true
104	City	Sedan	₹2200.0	✗ false
105	Nexon	SUV	₹1800.0	✓ true

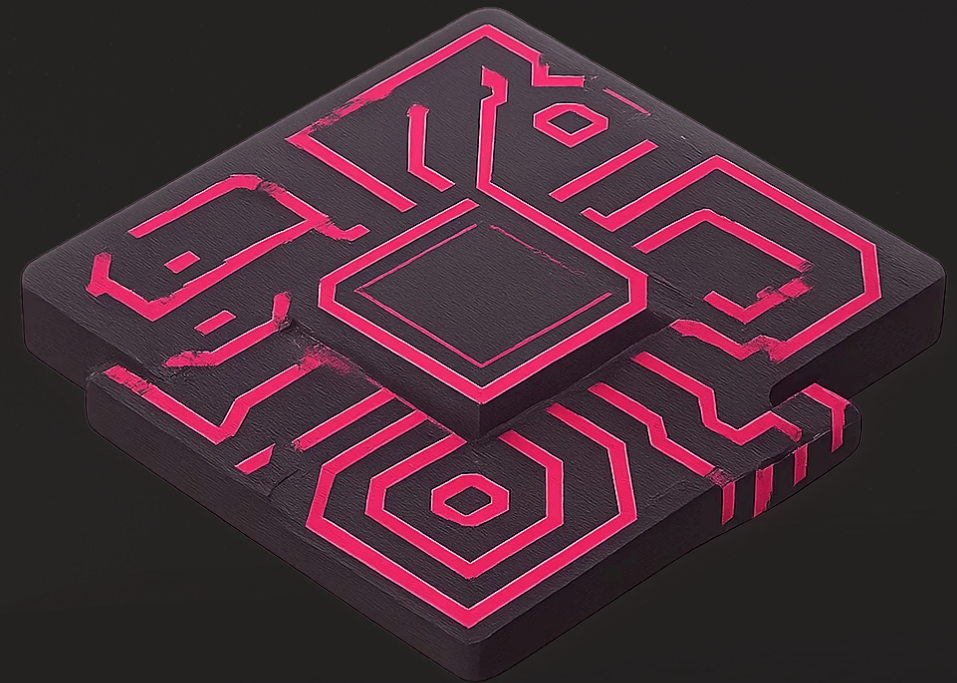
Fleet managed using **ArrayList** and **HashMap** (Module 6) – sorted by price via **Comparator** (Module 5) and filtered using **Streams** (Module 6).

Conclusion & Future Work

Conclusion

- Successfully demonstrates all six Java PSPJ modules in one cohesive project.
- Progresses from basic I/O and control flow to advanced OOP and functional programming.
- Integrates file handling, collections, and stream processing practically.
- This project deepened my understanding of Java fundamentals and OOP principles while helping me grow more confident in building structured, maintainable applications.
- It also strengthened my problem-solving and system design skills, improved my code organization and best practices, and gave me hands-on experience connecting multiple concepts into one cohesive solution.

Future Work



- GUI (Swing/JavaFX)
- Database Integration (MySQL/PostgreSQL)
- Web & REST APIs
- Mobile Application (Android/iOS)
- Advanced Features & Analytics
- Cloud Deployment & DevOps

Thank You

This Car Rental Fleet Manager project provided a comprehensive hands-on experience in applying core Java principles to a practical, real-world scenario. It effectively demonstrated object-oriented programming, data structures (like ArrayList and HashMap), file handling, and fundamental algorithms, all integrated into a functional application. We hope this presentation showcased the foundational knowledge and problem-solving skills acquired through this project.

"Programming is not about writing code - it's about solving problems systematically."

[View Project on GitHub](#)

[Click here](#)

Presented by: Gundu Meghana